

CATBALOGAN, Philippines

WMO: 985480

Lat: 11.7751N

Lon: 124.8842E

Elev: 16

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	72.4	73.7	68.0	103.1	76.1	69.5	108.6	77.4	15.8	79.6	10.7	80.7	1.2	40	0.319

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	12.3	93.6	81.2	92.7	81.0	91.8	80.9	83.1	90.6	82.5	90.1	81.9	89.3	5.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.0	161.1	88.0	80.4	157.8	87.4	79.9	155.5	86.9	47.1	91.0	46.3	90.4	45.8	90.0	88.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.5	9.3	7.7	DB	66.7	96.6	4.5	3.1	63.5	98.8	60.8	100.7	58.3	102.4	55.0	104.7	(4)
(5)			WB	65.7	85.9	3.8	1.8	62.9	87.2	60.7	88.2	58.5	89.3	55.7	90.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	82.8	80.0	80.7	82.1	83.8	84.9	84.3	83.7	84.5	83.8	83.1	82.1	81.2	(6)
(7)		DBStd	2.63	2.21	2.27	2.16	2.26	1.89	2.13	2.09	2.26	2.07	1.94	2.11	2.19	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	11988	930	859	994	1013	1083	1028	1045	1068	1013	1026	962	967	(10)
(11)		CDD65	6513	465	439	529	563	618	578	580	603	563	561	512	502	(11)
(12)		CDH74	71801	4006	4039	5454	6598	7719	6946	6805	7379	6641	6165	5248	4801	(12)
(13)		CDH80	26760	983	1211	1889	2705	3436	2889	2684	3173	2623	2192	1653	1322	(13)
(14)	Wind	WSAvg	2.8	2.4	2.5	2.7	2.6	2.9	2.8	3.2	3.5	3.5	2.7	2.2	2.8	(14)
(15)	Precipitation	PrecAvg	100.6	8.3	5.1	4.5	4.0	7.0	8.0	9.8	9.8	9.1	11.5	11.8	11.6	(15)
(16)		PrecMax	134.6	25.4	9.4	15.4	8.5	14.5	13.6	16.3	25.0	20.4	23.5	22.2	22.6	(16)
(17)		PrecMin	65.4	0.4	0.4	0.1	1.3	2.2	3.3	3.3	4.0	4.1	4.1	6.5	5.5	(17)
(18)		PrecStd	18.1	7.0	2.9	4.4	2.4	4.0	3.7	3.7	6.1	5.0	6.0	4.6	5.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.0	91.8	93.1	94.9	95.0	94.3	93.1	93.0	92.7	92.6	92.1	90.8	(19)
(20)			MCWB	78.8	79.2	79.3	80.2	81.8	81.9	81.5	81.4	81.8	80.9	80.8	80.3	(20)
(21)		2%	DB	88.3	89.9	91.2	93.3	93.5	92.8	91.6	91.8	91.4	91.2	90.6	89.4	(21)
(22)			MCWB	78.9	78.9	79.2	80.4	81.4	81.8	81.2	81.0	81.2	80.8	80.5	80.1	(22)
(23)		5%	DB	86.8	88.3	89.8	91.9	92.3	91.5	90.4	90.8	90.3	89.8	89.0	88.0	(23)
(24)			MCWB	78.7	78.6	79.2	80.4	81.3	81.5	80.9	80.8	80.9	80.7	80.2	79.9	(24)
(25)		10%	DB	85.2	86.6	88.3	90.4	91.0	90.1	89.1	89.7	89.1	88.3	87.3	86.3	(25)
(26)			MCWB	78.3	78.3	79.0	80.3	81.2	81.2	80.6	80.7	80.6	80.4	80.0	79.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.1	81.6	82.0	83.3	84.1	83.6	82.9	83.4	83.1	82.6	82.4	82.2	(27)
(28)			MCDWB	86.9	89.0	89.5	91.4	92.0	91.4	90.3	90.4	90.3	89.6	89.3	88.1	(28)
(29)		2%	WB	80.1	80.5	81.0	82.0	82.8	82.6	82.0	82.2	82.0	81.8	81.5	81.1	(29)
(30)			MCDWB	85.8	87.2	88.2	89.9	90.7	90.6	89.1	89.5	89.2	88.6	88.0	86.9	(30)
(31)		5%	WB	79.5	79.7	80.3	81.4	82.1	81.9	81.5	81.6	81.5	81.2	80.9	80.5	(31)
(32)			MCDWB	84.9	86.0	87.2	89.3	89.9	89.5	88.4	88.7	88.4	87.7	87.2	86.4	(32)
(33)		10%	WB	78.7	78.8	79.5	80.8	81.6	81.4	80.9	81.0	80.9	80.7	80.3	79.8	(33)
(34)			MCDWB	83.8	84.9	86.2	88.5	89.1	88.6	87.5	87.8	87.4	87.0	86.2	85.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.8	11.7	12.4	13.0	12.3	11.5	10.6	10.3	10.3	10.9	10.9	9.8	(35)
(36)			MCDBR	12.6	14.7	14.8	15.6	14.1	13.3	12.5	11.5	12.1	12.8	13.5	12.6	(36)
(37)			MCWBR	5.7	6.5	5.9	5.7	5.5	5.3	5.0	4.5	4.8	5.2	5.7	5.6	(37)
(38)		5% WB	MCDBR	11.0	12.6	13.0	13.8	13.0	12.6	11.7	10.9	11.2	11.6	11.9	11.2	(38)
(39)			MCWBR	5.8	6.6	6.2	6.1	5.8	5.6	5.3	5.0	5.1	5.3	5.8	5.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.458	0.436	0.432	0.425	0.424	0.423	0.426	0.448	0.439	0.447	0.440	0.457	(40)	
(41)		taud	2.298	2.364	2.390	2.408	2.434	2.454	2.444	2.365	2.389	2.366	2.386	2.308	(41)	
(42)		Ebn at Noon	268	279	282	281	276	273	273	271	276	273	272	265	(42)	
(43)		Edh at Noon	41	40	40	39	37	36	36	40	39	39	37	40	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1221	1511	1733	1906	1843	1657	1546	1616	1566	1488	1365	1207	(44)	
(45)		RadStd	171	192	193	131	144	94	119	96	140	104	122	111	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page